DP Biology – B3.2 (HL, Plant Systems)

Root pressure and phloem

- Which adaptations help animals and plants move fluids around?
- How does transport in animals compare with transport in plants?

HL statements B3.2.17–18. Textbook pp. 321–325.



Image: part of a sunflower stem, stained section. Jon Houseman, CC BY-SA 4.0.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	B3.2.17–18	Explain how root pressure is generated, and how sieve tubes and companion cells are adapted
2	B3.2.18	Explain translocation from source to sink, and evaluate the evidence for it

Root pressure

B3.2.17



Guttation: drops pushed out of a leaf by root pressure. Public domain.

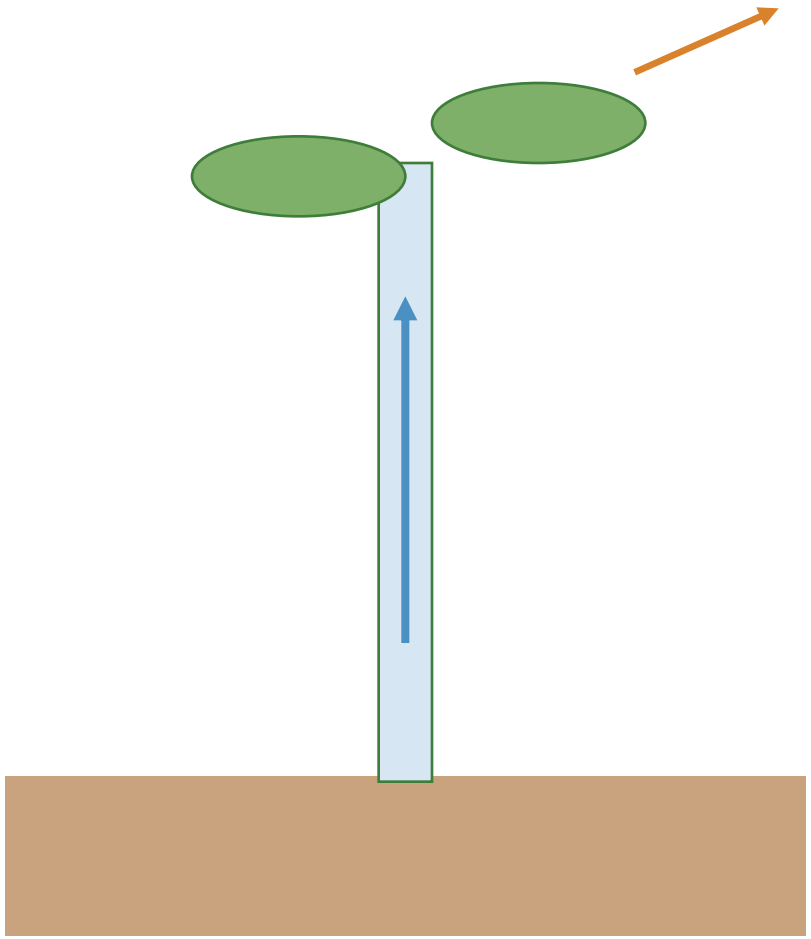
Usually water is **pulled** up by transpiration. When transpiration is low, roots can **push** sap up instead.

This happens in humid air, at night, or in spring before leaves open.

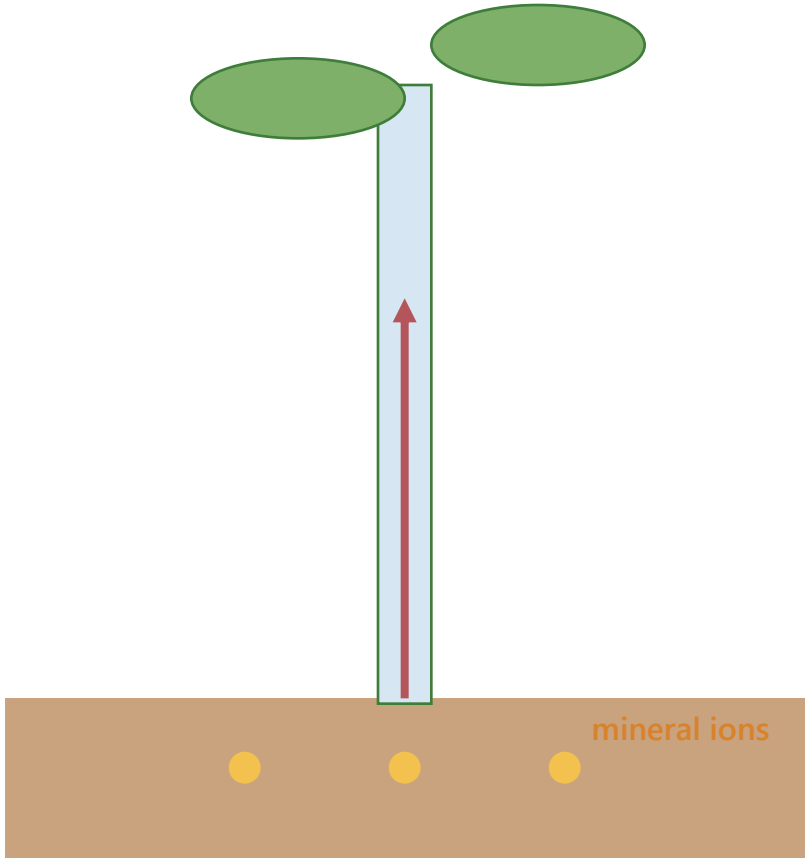
Root cells pump mineral ions into the xylem (**active transport**) → Xylem sap becomes more concentrated → Water follows by **osmosis** → Positive pressure pushes sap up

Pushed from below, pulled from above

B3.2.17



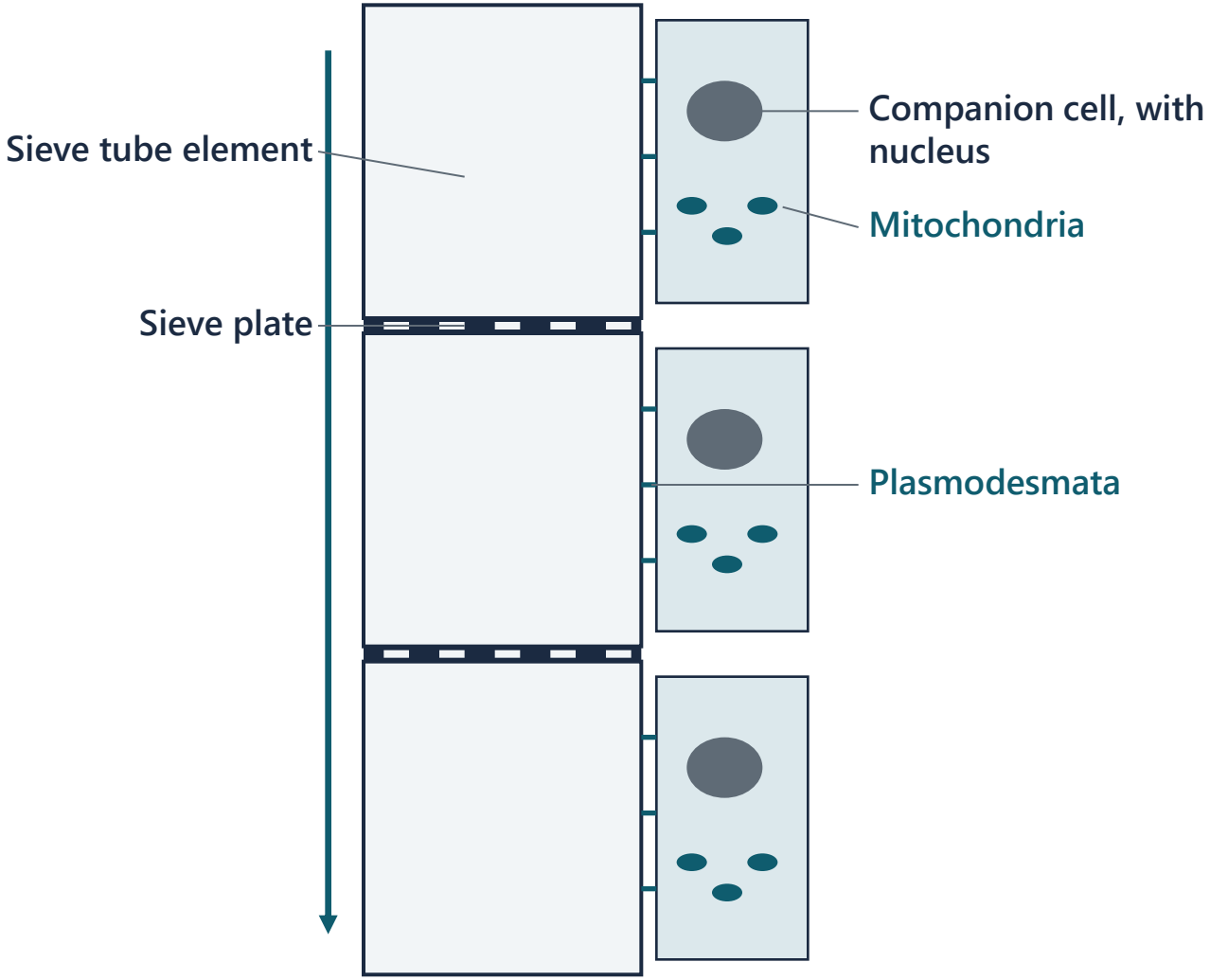
Day: tension pulls (transpiration)



Spring or night: root pressure pushes

Sieve tubes and companion cells

B3.2.18



Drawn longitudinal section. The arrow shows sap flow, which can go either way.

Translocation moves sap with sugars and amino acids from **sources** to **sinks**.

Sieve tube elements form the tube. Each has a **companion cell** that keeps it alive.

Adaptations for translocation

B3.2.18

Feature	How it helps
Sieve plates with large pores	Sap flows from one element to the next
Little cytoplasm, no nucleus	Less resistance to flow
Thick, rigid walls	Withstand high sap pressure
Many mitochondria in companion cells	ATP for loading sucrose
Plasmodesmata between the cells	Sucrose passes into the sieve tube

Key idea

Living tubes moving sap under pressure, unlike the dead xylem under tension.

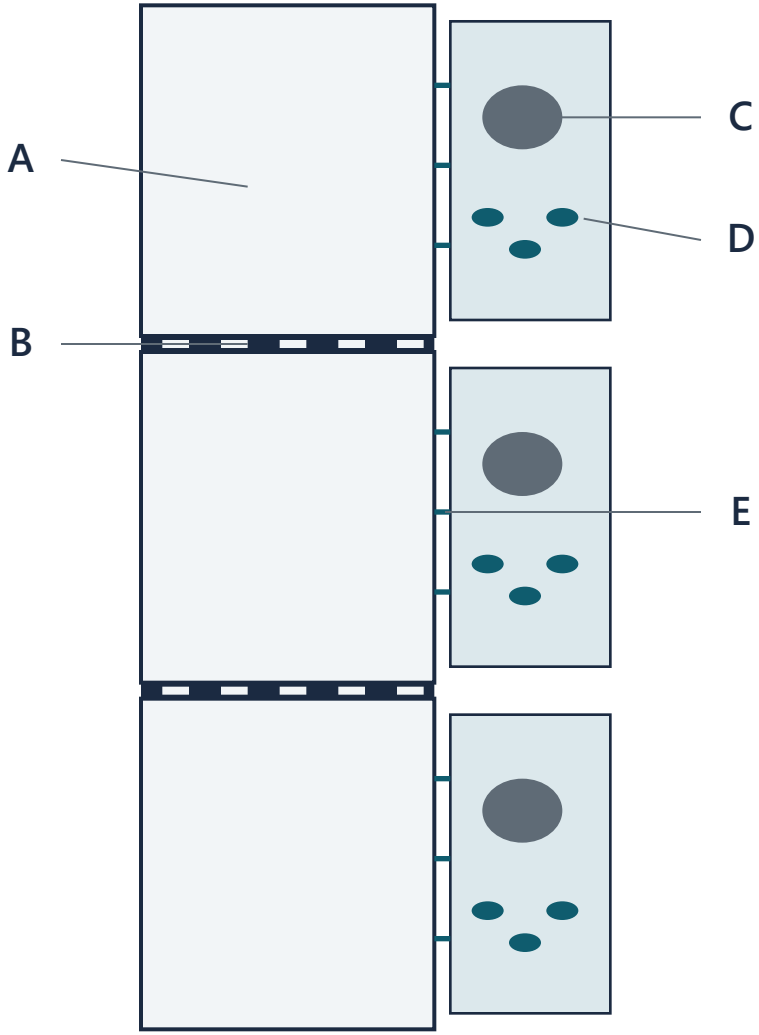
Xylem and phloem compared

B3.2.18

	Xylem	Phloem
Cells	Dead vessels	Living sieve tubes, companion cells
Carries	Water and mineral ions	Sugars and amino acids
Direction	Up only	Source to sink: up or down
Driven by	Tension (or root pressure)	High pressure from osmosis
End walls	None	Sieve plates

Link
Xylem: SL B3.2.7–8.

Practice: label the phloem



1. Name structures A to E.
2. Which structure has the nucleus?
3. Explain why D is numerous.
4. Explain why B has large pores.

Working with data: root pressure

B3.2.17

Sap pushed out of cut tomato stems in 1 hour. The values are invented for practice; the pattern is realistic.

Treatment of the roots	Sap / cm^3
Normal, well watered	1.6
Cyanide (stops respiration)	0.1
Cold, 4 °C	0.4
No oxygen	0.2

1. Explain the effect of cyanide.
2. Explain the effect of no oxygen.
3. Explain the result at 4 °C.
4. What do the results show about how root pressure is made?

When does root pressure matter?

B3.2.17

Spring morning

A maple has no leaves yet.

Humid night

Air is saturated with water vapour.

Hot afternoon

Strong sun and a dry wind.

For each one

Decide whether water rises mainly by root pressure or by transpiration pull, and explain why.

Root pressure and translocation side by side

B3.2.17–18

	Root pressure	Translocation
Tissue	Xylem	Phloem
What is moved	Water and ions	Sugars, amino acids
Energy	ATP to pump ions into xylem	ATP to load sucrose
Water moves by	Osmosis into xylem	Osmosis into sieve tubes

Key idea

Both use active transport to draw water in by osmosis, raising pressure.

Check-in: class 1

B3.2.17–18

1. Explain how root pressure is generated.
2. State two conditions in which root pressure matters most.
3. Describe the structure of a sieve tube element.
4. State two features of companion cells and why each helps.
5. Give three differences between xylem and phloem.

How to use this

Answer without your notes, then check against slides 3–11.

Sources and sinks

B3.2.18

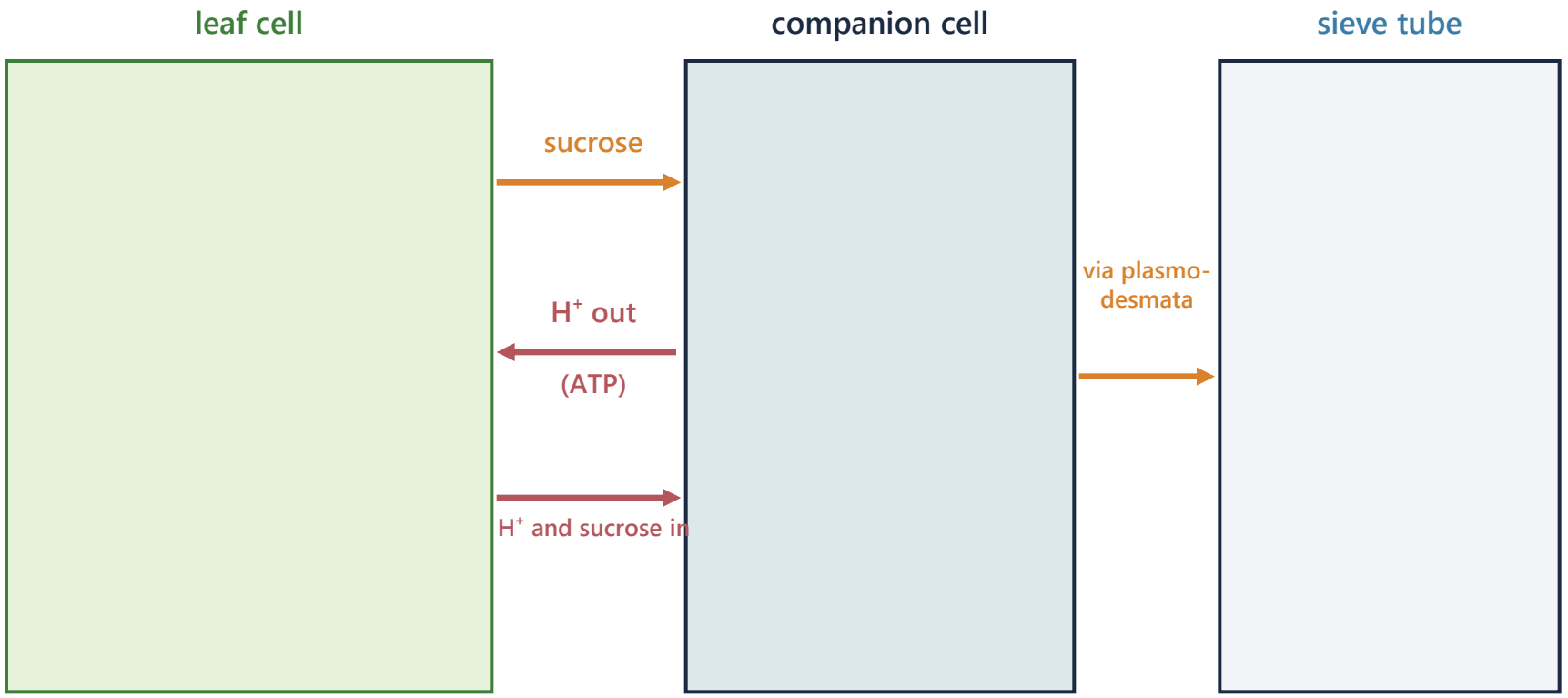
Organ	As a source	As a sink
Mature leaf	Photosynthesising in summer	Only when young
Root or tuber	In spring, feeding new shoots	In late summer, storing starch
Developing fruit	Never	While growing and ripening

Key idea

A **source** makes or releases sugar; a **sink** uses or stores it. The same organ can switch role, so phloem sap can flow up or down.

Loading sucrose at the source

B3.2.18



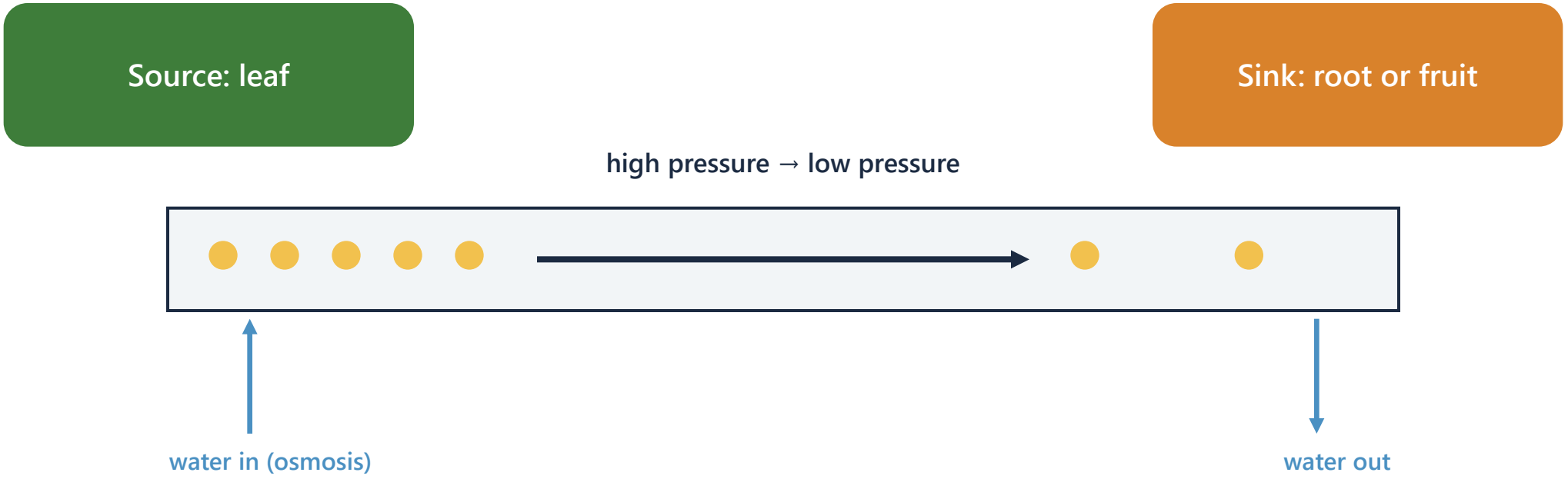
Drawn diagram, not to scale.

Companion cells pump H^+ out using **ATP**.

H^+ flows back in, carrying sucrose with it: **cotransport**.

Sucrose passes into the sieve tube.

Pressure moves the sap



At the source, sugar loading draws in water by osmosis: **high pressure**. At the sink, sugar is removed and water leaves: **low pressure**.

Sap flows down this pressure gradient by **mass flow**.

Unloading at the sink

B3.2.18



Turning sucrose into starch keeps the sink's sugar concentration low, so sucrose keeps diffusing in. Starch is insoluble and does not draw in water.

Think

Why does a ripening fruit taste sweeter, and what does this mean for the pressure gradient?

Evidence for translocation

B3.2.18

Aphid stylets

Cut off a feeding aphid: sap keeps oozing out under pressure, and can be analysed.

Radioactive tracers

Labelled CO_2 given to a leaf: labelled sugar appears in sinks.

Ringing

Remove a ring of bark: sugars build up above the ring.

Think

Why does sap flow out of a cut aphid stylet, and what does this tell us about phloem?

Working with data: a tracer experiment

B3.2.18

Radioactivity in parts of a bean plant 6 hours after one leaf was given labelled carbon dioxide. These data are invented for practice.

Part of plant	Radioactivity / %
Labelled leaf	48
Developing pods	30
Roots	14
Stem	6
Other mature leaves	2

1. Identify the main sinks.
2. Explain why other mature leaves have so little.
3. Explain why the stem has some.
4. Predict the result if the stem were ringed below the labelled leaf.

Practice: predict the flow

B3.2.18

Potato in spring

Tuber sprouting; no leaves yet.

Apple tree in summer

Leaves photosynthesising, fruits growing.

Aphids on a stem

Many aphids feeding on one shoot.

For each one

Name the source and the sink, and the direction of phloem flow. For the aphids, predict the effect on growth of the shoot.

Check-in: class 2

B3.2.18

1. Define a source and a sink, with an example of each.
2. Explain how sucrose is loaded into sieve tubes.
3. Explain how a pressure gradient forms between source and sink.
4. Describe one piece of evidence for translocation.
5. Explain why phloem can carry sap up or down.

How to use this

Answer without your notes, then check against slides 13–19.

Common mistakes to avoid

Not this	But this
Phloem only moves sap downwards.	It moves sap from source to sink, either way.
Sieve tubes have a nucleus.	Companion cells have the nucleus.
Root pressure lifts water in tall trees.	Transpiration pull does most of the work.
Phloem sap moves by diffusion.	It moves by mass flow under pressure.

Summary: pushing water, moving sugar

B3.2.17–18

	Key idea
B3.2.17	Ions pumped into xylem; water follows; pressure pushes sap up when transpiration is low
B3.2.18	Sieve tubes and companion cells; loading at sources, unloading at sinks; mass flow

Big picture

Water rises in dead xylem, mostly pulled by transpiration. Sugar moves in living phloem, pushed by pressure from osmosis. Both depend on active transport somewhere.

Check that you can...

1. **Explain** how root pressure is generated and when it matters
2. **Describe** sieve tubes and companion cells
3. **Explain** how sucrose is loaded at a source
4. **Explain** translocation from source to sink
5. **Evaluate** evidence for the pressure-flow model

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Outline how active transport in root cells leads to root pressure.
2. State two adaptations of sieve tube elements for the flow of sap.
3. Explain why sap flows from a source to a sink.

Image credits

Helianthus stem.jpg – Jon Houseman – CC BY-SA 4.0

Guttation ne.jpg – NoahElhardt assumed (ba – Public domain

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